Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The straight lines joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P intersect at R. Then the equation of the locus of R is

(A) $x^2 + 2y^2 - ax = 0$

(B) $2x^2 + y^2 - 2ax = 0$

(C) $2x^2 + 2y^2 - ay = 0$

(D) $2x^2 + y^2 - 2ay = 0$

Q.2

The directrix of the parabola $y^2 + 4x + 3 = 0$ is

(A) $x - 3/4 = 0$

(B) $x + 1/4 = 0$

(C) $x - 1/4 = 0$

(D) $x - 4/3 = 0$

Q.3

The tangent at any point P on the parabola $y^2 = 4ax$ intersects the y-axis at Q. The tangent to the circumcircle of triangle PQS (S is the focus) at Q is

(A) a line parallel to x-axis

(B) y-axis

(C) a line parallel to y-axis

(D) none of these

Q.4

If $y = 2x - 3$ is a tangent to the parabola $y^2 = 4a\left(x - \frac{1}{3}\right)$ then a is equal to

(A) $\frac{22}{3}$

(B) -1

(C) $\frac{14}{3}$

(D) $\frac{-14}{3}$

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Q.5

The length of the perpendiculars from the focus and the extremities of a focal chord of a parabola on the tangent at the vertex form

(A) an A.P.

(B) a G.P.

(C) an H.P.

(D) none of these.

Q.6

For the parabola $y^2 + 8x - 12y + 20 = 0$, which of the following in not correct

(A) vertex $(2, 6)$

(B) focus $(0, 6)$

(C) length of the latus rectum $= 4$

(D) axis is $y = 6$

Q.7

If the tangent to the conic $y - 6 = x^2$ at $(2, 10)$ touches the circle $x^2 + y^2 + 8x - 2y = k$ (for same fixed k) at a point (α, β); then (α, β) is:

(A) $\left(-\frac{6}{17}, \frac{10}{17}\right)$

(B) $\left(-\frac{8}{17}, \frac{2}{17}\right)$

(C) $\left(-\frac{4}{17}, \frac{1}{17}\right)$

(D) $\left(-\frac{7}{17}, \frac{6}{17}\right)$

Q.8

An equation of a common tangent to the parabola $y^2 = 4x$ and $x^2 = 4y$ is

(A) $x - y + 1 = 0$

(B) $x + y - 1 = 0$

(C) $x + y + 1 = 0$

(D) $y = 0$

Q.9

If P be the point $(1, 0)$ and Q a point on the locus $y^2 = 8x$. The locus of the mid point of PQ is

(A) $x^2 + 4y + 2 = 0$

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(B) $x^2 - 4y + 2 = 0$

(C) $y^2 - 4x + 2 = 0$

(D) $y^2 + 4x + 2 = 0$

Q.10

The tangents to $y^2 = 4ax$ make angles θ_1 and θ_2 with the x-axis. If $\cos\theta_1 \cos\theta_2 = \lambda$, then the locus of their point of intersection is

(A) $x^2 = \lambda^2 \left[(x-a)^2 + 4y^2\right]$

(B) $x^2 = \lambda^2 \left[(x+a)^2 + y^2\right]$

(C) $x^2 = \lambda^2 \left[(x-a)^2 + y^2\right]$

(D) $4x^2 = \lambda^2 \left[(x+a)^2 + y^2\right]$

Q.11

If two tangents drawn from a point P to the parabola $y^2 = 4x$ are at right angles, then the locus of P is

(A) $x = -1$

(B) $2x - 1 = 0$

(C) $x = 1$

(D) $2x + 1 = 0$

Q.12

If $2x + y + \lambda = 0$ is a normal to the parabola $y^2 = -8x$, then λ is

(A) 12

(B) -12

(C) 24

(D) -24

Q.13

If P is the point $(1, 0)$ and Q lies on the parabola $y^2 = 36x$, then the locus of the mid point of PQ is:

(A) $y^2 = 9(2x - 1)$

(B) $y^2 = 9(x + 2)$

(C) $y^2 = 2(x - 9)$

(D) $x^2 = 9(y - 2)$

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Q.14

Equation of the tangent at a point P on the parabola $y^2 = 4ax$, the normal at which is at a distance $a\sqrt{5}/4$ from the focus of the parabola is

(A) $4x - 2y + a = 0$

(B) $4x - 8y + 9a = 0$

(C) $2x - y - 12a = 0$

(D) $2x + y - 12a = 0$

Q.15

If the normal to the parabola $y^2 = 4ax$ at P meets the curve again at Q and if PQ and the normal at Q make angle α and β, respectively, with the x-axis, then $\tan\alpha(\tan\alpha + \tan\beta)$ has the value equal to

(A) 0

(B) -2

(C) $-1/2$

(D) -1

Q.16

A chord is drawn through the focus of the parabola $y^2 = 6x$ such that its distance from the vertex of this parabola is $\frac{\sqrt{5}}{2}$, then its slope can be:

(A) $\frac{\sqrt{5}}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\frac{2}{\sqrt{5}}$

(D) $\frac{2}{\sqrt{3}}$

Q.17

Locus of the mid points of the chords of the parabola $y^2 = 8x$ which touch the circle $x^2 + y^2 = 4$ is

(A) $\left(y^2 - 4x\right)^2 = 16\left(x^2 + 4\right)$

(B) $\left(y^2 - 4x^2\right) = 4\left(x^2 + a^2\right)$

(C) $\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

(D) $\left(y^2 - 4x\right)^2 = 16\left(4 + y^2\right)$

Q.18

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If a tangent to the parabola $y^2 = 4x$ makes an angle $\pi/4$ with the positive direction of the axis of x then the coordinates of the point of contact with the parabola are:

(A) $(1, 2)$

(B) $(1, -2)$

(C) $(4, 4)$

(D) $(4, -4)$

Q.19

Through the vertex O of the parabola $y^2 = 4ax$, two chords OP and OQ are drawn and the circles on OP and OQ as diameters intersect at R. If θ_1, θ_2, and ϕ are the angles made with the axis by the tangents at P and Q on the parabola and by OR, then the value of $\cot\theta_1 + \cot\theta_2$ is

(A) $-2\tan\phi$

(B) $-2\tan(\pi - \phi)$

(C) 0

(D) $2\cot\phi$

Q.20

If the normal at P to the parabola $y^2 = 4ax$ meets the parabola again at Q such that PQ subtends a right angle at the vertex of the parabola, then the coordinates of P are

(A) $(2a, 2\sqrt{2}a)$

(B) $(-2a, 2\sqrt{2}a)$

(C) $(2\sqrt{2}a, 2a)$

(D) $(2\sqrt{2}a, -2a)$

Q.21

PQ is any focal chord of the parabola $y^2 = 8x$. Then the length of PQ can never be less than

Q.22

If $3x + 4y + k = 0$ represents the equation of tangent at the vertex of the parabola $16x^2 - 24xy + 9y^2 + 14x + 2y + 7 = 0$, then the value of k is

Q.23

If the line $x + y = 6$ is a normal to the parabola $y^2 = 8x$ at point (a, b), then the value of $a + b$ is

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Q.24

Normals are drawn from the point P with slopes m_1, m_2 and m_3 to that parabola $y^2 = 4x$. If the locus of P with $m_1 m_2 = \alpha$ is a part of the parabola itself then the value of α is

Q.25

The focal chord of $y^2 = 16x$ is tangent to $(x-6)^2 + y^2 = 2$ Then the possible value of the square of slope of this chord is

Q.26

The equation of the line touching both the parabolas $y^2 = 4x$ and $x^2 = -32y$ is $ax + by + c = 0$. Then the value of $a+ b + c$ is

Q.27

Points A, B and C lie on the parabola $y^2 = 4ax$. The tangents to the parabola at A, B and C, taken in pairs, intersect at points P, Q and R. Then the ratio of the areas of the triangles ABC and PQR is

Q.28

Foot of perpendicular from point P on the parabola $y^2 = 4ax$ to the axis is N. A straight line is drawn parallel to the axis which bisects PN and cuts the curve at Q. If NQ meets the tangent at the vertex A at a point T, then $\frac{PN}{AT} =$

Q.29

If the locus of the middle of point of contact of tangent drawn to the parabola $y^2 = 8x$ and the foot of perpendicular drawn from its focus to the tangents is a conic, then the length of latus rectum of this conic is

Q.30

From the point $(-1, 2)$, tangent lines are drawn to the parabola $y^2 = 4x$. If the area of the triangle formed by the chord of contact and the tangents is A, then the value of $A/\sqrt{2}$ is

Answers & Solutions

Q.1 Answer:

$2x^2 + y^2 - 2ax = 0$

Solution:

Given the equation of parabola is

$y^2 = 4ax$

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Let $\mathrm{P}\left(\mathrm{at}^2, 2\mathrm{at}\right)$ be any point on the parabola.

Equation of tangent at point P is $\mathrm{ty} = \mathrm{x} + \mathrm{at}^2$ where slope of the tangent is $\frac{1}{\mathrm{t}}$.

Equation of line perpendicular to the tangent passes through $(\mathrm{a}, 0)$ is given as $\therefore \mathrm{y} - 0 = -\mathrm{t}(\mathrm{x} - \mathrm{a})$

or $\mathrm{y} = \mathrm{t}(\mathrm{a} - \mathrm{x})$...(i)

Equation of OP is given by

$\mathrm{y} - 0 = \frac{2}{\mathrm{t}}(\mathrm{x} - 0) = 0$

$\Rightarrow \mathrm{y} = \frac{2}{\mathrm{t}}\mathrm{x}$...(ii)

Eliminating 't' from equations (i) and (ii), we get

$\mathrm{y}^2 = 2\mathrm{x}(\mathrm{a} - \mathrm{x})$

Q.2 Answer:

$x - 1/4 = 0$

Solution:

The parabola is, $y^2 + 4x + 3 = 0$

$\Rightarrow y^2 = -4x - 3 = -4\left(x + \frac{3}{4}\right)$

Now comparing above equation with standard parabola, $\mathrm{Y}^2 = -4\mathrm{aX}$

$\mathrm{Y} = \mathrm{y}, \mathrm{a} = 1, \mathrm{X} = \mathrm{x} + \frac{3}{4}$

Hence directrix is $\mathrm{X} = \mathrm{a}$

$\Rightarrow \mathrm{x} + \frac{3}{4} = 1$

$\Rightarrow \mathrm{x} - \frac{1}{4} = 0$

Q.3 Answer:

y-axis

Solution:

Given equation of parabola is

$\mathrm{y}^2 = 4\mathrm{ax}$

Let $\mathrm{P}\left(\mathrm{at}^2, 2\mathrm{at}\right)$ be any point on the parabola.

Equation of tangent at P is

$\mathrm{ty} = \mathrm{x} + \mathrm{at}^2$

Since the tangent meets y-axis at Q i.e. $\mathrm{x} = 0$

$\Rightarrow \mathrm{y} = \mathrm{at}$

So, the coordinates of Q is $(0, \mathrm{at})$.

So, we draw the circumcircle of triangle PQS.

So, from the graph, its clear that y-axis is the tangent to the circumcircle of triangle PQS at Q.

Q.4 Answer:

$\frac{-14}{3}$

Solution:

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Solving $y = 2x - 3$ and $y^2 = 4a\left(x - \frac{1}{3}\right)$, we have

$(2x-3)^2 = 4a\left(x - \frac{1}{3}\right)$

$\Rightarrow 4x^2 + 9 - 12x = 4ax - \frac{4a}{3}$

$\Rightarrow 4x^2 - 4(3+a)x + 9 + \frac{4a}{3} = 0$

This equation must have equal roots $\Rightarrow D = 0$

$\Rightarrow 16(3+a)^2 - 16\left(9 + \frac{4a}{3}\right) = 0$

$\Rightarrow 9 + a^2 + 6a = 9 + \frac{4a}{3}$

$\Rightarrow a^2 + \frac{14a}{3} = 0$

$\Rightarrow a = 0$ or $a = -\frac{14}{3}$

Q.5 Answer:

a G.P.

Solution:

$y^2 = 4ax$ be the parabola, $P\left(at^2, 2at\right), Q\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$ and the focus is $S(a, 0)$. The tangent at the vertex is $x = 0$. Lengths of the perpendicular from P, Q and S on this tangent are respectively $l_1 = at^2\ l_2 = a/t^2, l_3 = a$ so that $l_1 l_2 = a^2 = l_3^2 \Rightarrow l_1, l_2\ l_3$ form a G.P.

Q.6 Answer:

length of the latus rectum $= 4$

Solution:

$(y-6)^2 = -8(x-2)$ or $Y^2 = 4AX$ where $Y = y - 6\ X = x - 2, A = -2$, vertex is $(2, 6)$, focus is $(X = A, Y = 0)$ i.e., $(0, 6)$ axis is $Y = 0$ i.e. $y = 6$ and length of the latus rectum $= |4A| = 8$

Q.7 Answer:

$\left(-\frac{8}{17}, \frac{2}{17}\right)$

Solution:

An equation of tangent to

$y - 6 = x^2$ **at** $(2, 10)$ **is**

$(y + 10) - 12 = 2(2)x$ or $4x - y + 2 = 0$ **(1)**

An equation of normal at the point of contact (α, β) of (1) and $x^2 + y^2 + 8x - 2y = k$ is

$(x - (-4)) + 4(y - 1) = 0$

or $x + 4y = 0$ **(2)**

The point (α, β) is the point of intersection of (1) and (2). Solving (1) and (2), we get

$\alpha = -\frac{8}{17}, \beta = \frac{2}{17}$

Q.8 Answer:

$x + y + 1 = 0$

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Solution:

Equation of a tangent to $y^2 = 4x$ is $y = mx + 1/m$ If it touches the parabola $x^2 = 4y$ then the roots of the equation $x^2 = 4(mx + 1/m)$ are coincident. So $(4m)^2 + \frac{16}{m} = 0 \Rightarrow m = -1$ and the equation of the tangent is $x + y + 1 = 0$

Q.9 Answer:

$y^2 - 4x + 2 = 0$

Solution:

Let Q be $\left(2t^2, 4t\right)$, $P(1, 0)$

Mid point of $PQ = (h, k) = \left(\frac{2t^2+1}{2}, 2t\right)$

Eliminating t we get

$h = \frac{2\left(\frac{k}{2}\right)^2+1}{2} \Rightarrow 4h = k^2 + 2$

Locus of (h, k) is $y^2 - 4x + 2 = 0$

Q.10 Answer:

$x^2 = \lambda^2 \left[(x - a)^2 + y^2\right]$

Solution:

equation of tangent in slope form is $\mathrm{m^2x - ym + a = 0}$

Therefore, $\mathrm{m_1 + m_2 = \tan\theta_1 + \tan\theta_2 = \frac{y}{x}}$...(1)

and $\mathrm{m_1\ m_2 = \tan\theta_1 \tan\theta_2 = \frac{a}{x}}$...(2)

now, $\mathrm{\cos\theta_1 \cos\theta_2 = k}$

$\Rightarrow \mathrm{\sec^2\theta_1 \sec^2\theta_2 = \frac{1}{k^2}}$

$\Rightarrow \left(1 + \tan^2\theta_1\right)\left(1 + \tan^2\theta_2\right) = \frac{1}{k^2} 1 + \left(\tan\theta_1 + \tan\theta_2\right)^2 - 2\tan\theta_1 \tan\theta_2 +$

$\tan^2\theta_1 \tan^2\theta_2 = \frac{1}{k^2}$ from (1) and $(2) : 1 + \frac{y^2}{x^2} - \frac{2a}{x} + \frac{4a^2}{x^2} = \frac{1}{k^2}\, x^2 = k^2 \left[(x - a)^2 + y^2\right]$

Q.11 Answer:

$x = -1$

Solution:

Given equation of parabola is $y^2 = 4x$..... (i)

We know that the locus of point P from which two perpendicular tangents are drawn to the parabola is the directrix of the parabola.

The standard equation of parabola $\mathrm{(y - k)^2 = 4p(x - h)}$ has focus $\mathrm{(h + p, k)}$ and the directrix is $\mathrm{x = h - p}$

From (i), we get

$\mathrm{h = 0, k = 0, p = 1}$

$\Rightarrow$ Directrix of (i) is $\mathrm{x = 0 - 1 = -1}$

Hence, required locus is $\mathrm{x = -1}$.

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Q.12 Answer:

24

Solution:

Given the equation of parabola is

$y^2 = -8x$

Here, $\mathbf{a} = 2$

Equation of normal to parabola having slope m **is**

$\mathrm{y} = \mathrm{mx} + 2\mathrm{am} + \mathrm{am}^3$

$\mathrm{y} = \mathrm{mx} + 4\ \mathrm{m} + 2\ \mathrm{m}^3$

$\text{or } \mathrm{y} - \mathrm{mx} - 4\ \mathrm{m} - 2\ \mathrm{m}^3 = 0$

Given the equation of normal to parabola is

$2x + y + \lambda = 0$

On comparing, we get

$\mathrm{m} = -2 \text{ and } \lambda = -4\ \mathrm{m} - 2\ \mathrm{m}^3$

$\Rightarrow \lambda = 8 + 16 = 24$

Q.13 Answer:

$y^2 = 9(2x - 1)$

Solution:

Let the coordinates of Q **be** $\left(9t^2, 18t\right)$ **coordinates of** R **the mid point of** PQ **are** $\left(\frac{9t^2+1}{2}, 9t\right) = (x, y)$**. Locus of** R **is**

$2x = 9\left(\frac{y}{9}\right)^2 + 1$

$\Rightarrow y^2 = 9(2x - 1)$

Q.14 Answer:

$4x - 2y + a = 0$

Solution:

Equation of the normal with slope m **is** $y = mx - 2am - am^3$ **whose distance from the focus** $(a, 0)$ **is** $\frac{am-2am-am^3}{\sqrt{1+m^2}} = \frac{\sqrt{5}}{4}a$

$\Rightarrow \quad 16m^2\left(1 + m^2\right) = 5$

$\Rightarrow m^2 = 1/4 \Rightarrow m = \pm 1/2$

Slope of the tangent is ± 2 **and the equation of the tangent is**

$y = \pm 2x \pm a/2 \Rightarrow 4x \pm 2y + a = 0$

Q.15 Answer:

-2

Solution:

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Let the given equation of parabola is

$y^2 = 4ax$

Let the normal at $\mathbf{P}\left(\mathbf{at_1}^2, \mathbf{2at_1}\right)$ meets the parabola again at $\mathrm{Q}\left(\mathrm{at_2}^2, \mathrm{2at_2}\right)$.

Equation of normal at P is given by,

$\mathbf{y = -t_1 x + 2at_1 + at_1^3}$

Equation of normal at Q is given by,

$\mathbf{y = -t_2 x + 2at_2 + at_2^3}$

So, $\tan\alpha = -\mathrm{t}_1$ and $\tan\beta = -\mathrm{t}_2$

Also, $t_2 = -t_1 - \frac{2}{t_1}$

$\Rightarrow \mathrm{t_1 t_2 + t_1^2 = -2}$

$\Rightarrow \tan\alpha\tan\beta + \tan^2\alpha = -2$

Q.16 Answer:

$\frac{\sqrt{5}}{2}$

Solution:

Let the equation of the chord be.

$y - 0 = m\left(x - \frac{3}{2}\right)$

Its distance from the vertex $(0,0)$ is $\left|\frac{m\left(-\frac{3}{2}\right)}{\sqrt{1+m^2}}\right| = \frac{\sqrt{5}}{2} \Rightarrow m = \pm\frac{\sqrt{5}}{2}$

Q.17 Answer:

$\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

Solution:

Let (h,k) be the mid-point of the chord of the parabola $y^2 = 8x$, then its equation is

$ky - 4(x+h) = k^2 - 8h$

$\Rightarrow 4x - ky + k^2 - 4h = 0$

which touches the circle $x^2 + y^2 = 4$

$\Rightarrow \frac{k^2 - 4h}{\sqrt{16+k^2}} = 2$

$\Rightarrow \left(k^2 - 4h\right)^2 = 4\left(k^2 + 16\right)$

Locus of (h,k) is $\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

Q.18 Answer:

$(1,2)$

Solution:

Slope of the tangent to the parabola at $\left(t^2, 2t\right)$ is $\frac{1}{t} = \tan\frac{\pi}{4} = 1 \Rightarrow t = 1$ and the required point of contact is $(1,2)$

Q.19 Answer:

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$-2\tan\phi$

Solution:

$x\left(x-at_1^2\right)+y\left(y-2at_1\right)=0$

$\mathrm{x}^2+\mathrm{y}^2-\mathrm{at}_1^2\mathrm{x}-2\mathrm{at}_1\mathrm{y}=0$

Equation of circle at Q:-

$\mathrm{x}^2+\mathrm{y}^2-\mathrm{at}_2^2\mathrm{x}-2\mathrm{at}_2\mathrm{y}=0$

Equation of common chord:-

$2\left(-\mathrm{at}_1^2+\mathrm{at}_2^2\right)\mathrm{x}+2\left(-2\mathrm{at}_1+2\mathrm{at}_2\right)\mathrm{y}=0$

$\left(\mathrm{t}_2^2-\mathrm{t}_1^2\right)\mathrm{x}+2\left(\mathrm{t}_2-\mathrm{t}_1\right)\mathrm{y}=0$

$\mathrm{y}=\frac{-(\mathrm{t}_2+\mathrm{t}_1)\mathrm{x}}{2}$

$\tan\phi=\frac{-(t_2+t_1)}{2}$

so, $\tan\phi=\frac{-(\cot\theta_1+\cot\theta_2)}{2}$

$\Rightarrow\cot\theta_1+\cot\theta_2=-2\tan\phi$

Q.20 Answer:

$(2a,2\sqrt{2}a)$

Solution:

Let the coordinates of P **be** $\left(am^2,-2am\right)$ **and** Q **be** $\left(am'^2,-2am'\right)$ **then** $\frac{-2am}{am^2}\times\frac{-2am'}{am'^2}=-1\Rightarrow\quad mm'=-4$

Normal at $P, y=mx-2am-am^3$**. Since it passes through** Q

$-2a\left(-\frac{4}{m}\right)=am\left(\frac{16}{m^2}\right)-2am-am^3=0$

$\Rightarrow m^4+2m^2-8=0\Rightarrow m=\pm\sqrt{2}$ **and the coordinates of** P **are** $(2a,\pm2\sqrt{2}a)$

Q.21 Answer:

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Solution:

The length of focal chord having one extremity $\left(at^2,2at\right)$ **is**

$a\left(r+\frac{1}{t}\right)^2$

$\because\left|r+\frac{1}{t}\right|\geq2$

we get $a\left(t+\frac{1}{t}\right)^2\geq4a=8$

or Length of focal chord 8

Q.22 Answer:

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Solution:

Given that equation of tangent at vertex is $3x+4y+k=0$ **Let the equation of axis be** $4x-3y+\lambda=0$ **Therefore, equation of parabola is**

$(4x-3y+\lambda)^2=\mu(3x+4y+k)$

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Given equation of parabola is

$16x^2 - 24xy + 9y^2 + 14x + 2y + 7 = 0$

Comparing equations (1) and (2), we get

$8\lambda - 3\mu = 14, -6\lambda - 4\mu = 2, \lambda^2 - k\mu = 7$

Solving, we get

$\lambda = 1, \mu = -2$ **and** $k = 3$

Q.23 Answer:

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Solution:

Slope of the normal $x + y = 6$ **is** $m_1 = -1$

Then the slope of the Tangent at the same point is $\mathrm{m} = \frac{-1}{\mathrm{m_1}} = 1$

Slope of the tangent for the given curve at point $(\mathrm{x_1}, \mathrm{y_1})$ **is given as**

$2\mathrm{y_1} \cdot \mathrm{m} = 8 \Rightarrow \mathrm{y_1} = 4 \Rightarrow \mathrm{x_1} = 2$

point is : $(2, 4)$

Q.24 Answer:

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Solution:

We know, equation of normal to $y^2 = 4ax$ **is**

$y = mx - 2am - am^3$

Thus, equation of normal to $y^2 = 4x$ **is,** $y = mx - 2m - m^3$

Let it passes through (h, k)

$\Rightarrow k = mh - 2m - m^3$

Or $\mathrm{m}^3 + \mathrm{m}(2 - \mathrm{h}) + \mathrm{k} = 0 \quad \ldots(\mathrm{i})$

Here, $m_1 + m_2 + m_3 = 0$,

$m_1m_2 + m_2m_3 + m_3m_1 = 2 - h$

$\mathrm{m_1\ m_2\ m_3} = -\mathrm{k}$ **Where** $\mathrm{m_1\ m_2} = \mathrm{a}$

$\Rightarrow \mathrm{m_3} = -\frac{\mathrm{k}}{\mathrm{a}}$ **it must satisfy Eq. (i)**

$\Rightarrow \frac{\mathrm{k}^3}{\mathrm{a}^3} - \frac{\mathrm{k}}{\mathrm{a}}(2 - \mathrm{h}) + \mathrm{k} = 0$

$\Rightarrow k^2 = a^2h - 2a^2 + a^2$

$\Rightarrow y^2 = \alpha^2x - 2a^2 + \alpha^3$

On comparing with $y^2 = 4x \Rightarrow a^2 = 4$ **and** $-2a^2 + a^3 = 0$

$\Rightarrow a = 2$

Q.25 Answer:

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Solution:

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$y^2 = 16x$

Here, $4a = 16 \Rightarrow a = 4$

Focus of the parabola (i) is $(4, 0)$

Focal chord of the parabola is tangent to the circle $(x-6)^2 + y^2 = 2$

$\sqrt{2}$ **and** $(6, 0)$ **are radius and centre of the circle**

As radius is perpendicular to the tangent, we have length of tangent from $(4, 0)$ **to the circle is** $= \sqrt{2}$.

From the diagram, we have

$\tan\theta = \frac{\sqrt{2}}{\sqrt{2}} = 1 \Rightarrow \theta = 45^\circ$

Therefore, slope of the chord is $\pm 1 = (-1, 1)$.

Q.26 Answer:

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Solution:

The equation of any tangent to $y^2 = 4x$ **is**

$y = mx + \frac{1}{m} \cdots$ **(i)**

If it touches the parabola $x^2 = -32y$**, then equation**

$x^2 = -32\left(mx + \frac{1}{m}\right)$

i.e. $x^2 + 32mx + \frac{32}{m} = 0 \Rightarrow xm^3 - 1 = 0 \Rightarrow m = \frac{1}{2}$

Substituting the value of m **in (i), we get** $x - 2y + 4 = 0$

Q.27 Answer:

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Solution:

Let the three points on the parabola $y^2 =$ **4ax be**

Then equations of tangents at A, B **and** c **are**

$y = \frac{x}{t_1} + at_1$ **(1)**

$y = \frac{x}{t_2} + a_2$ **(2)**

$y = \frac{x}{t_3} + at_3$ **(3)**

Solving the above equations pairwise, we get the points

$P\left(at_1t_2, a\left(t_1 + t_2\right)\right), Q\left(at_2t_3, a\left(t_2 + t_3\right)\right), R\left(at_3t_1, a\left(t_3 + t_1\right)\right)$

NOW area of $\triangle ABC = \frac{1}{2}\begin{vmatrix} 1 & at_3^2 & 2at_1 \\ 1 & at_2^2 & 2at_2 \\ 1 & at_3^2 & 2at_3 \end{vmatrix}$

$= \mathrm{a}^2\begin{vmatrix} 1 & \mathrm{ta}_1 & t_1^? \\ 1 & \mathrm{t}_2 & t_2^2 \\ 1 & t_3 & t_3^? \end{vmatrix}$

$= \left|a^2\left(t_1 - t_2\right)\left(t_2 - t_3\right)\left(t_3 - t_1\right)\right| \quad (4)$

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Also area Of $\triangle PQR = \frac{1}{2}\begin{vmatrix} 1 & a_1t_2 & a\left(t_1+t_2\right) \\ 1 & a_2t_3 & a\left(t_2+t_3\right) \\ 1 & a_3t_1 & a\left(t_3+t_1\right) \end{vmatrix}$

$$= \frac{a^2}{2}\begin{vmatrix} 1 & t_1t_2 & t_1+t_2 \\ 1 & t_2t_3 & t_2+t_3 \\ 1 & t_3t_1 & t_3+t_1 \end{vmatrix}$$

$$= \frac{a^2}{2}\begin{vmatrix} 0 & \left(t_1-t_3\right)t_2 & t_1-t_3 \\ 0 & \left(t_2-t_1\right)t_3 & t_2-t_1 \\ 1 & t_3t_1 & t_3+t_1 \end{vmatrix}$$

Expanding along c_1

$= \left| \frac{a^2}{2}\left(t_1-t_0\right)\left(t_2-t_1\right)\left(t_2-t_3\right| \right.$

From Eqs. (4) and (5), we get $\frac{\mathrm{Ar}(\triangle ABC)}{\mathrm{Ar}(AMBC)} = \frac{a^2|(t_1-t_2)(t_2-t_3)(t_3-t_1)|}{\frac{a^2}{2}|(t_1-t_2)(t_2-t_3)(t_3-t_1)|}$

$= \frac{2}{1}$

Therefore, the required ratio is $2:1$

Q.28 Answer:

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Solution:

Let point P **on the parabola be** $\left(at^2, 2at\right)$.

$\therefore \quad PN = 2at$ **and** $N \equiv \left(at^2, 0\right)$

M **is midpoint of** PN.

Therefore, equation of QM **is** $y = at$

Solving this with parabola, we have

$a^2t^2 = 4ax \Rightarrow x = \frac{1}{4}at^2$

So, $Q \equiv \left(\frac{1}{4}at^2, at\right)$

Thus, equation of NQ **is**

$y - 0 = \frac{at-0}{\frac{1}{4}at^2-at^2}\left(x-at^2\right)$

$y = -\frac{4}{3t}\left(x-at^2\right)$

If NQ meets y**-axis at** $T\left(0, \frac{4at}{3}\right)$**, then**

$AT = \frac{4at}{3}$

$\therefore \quad \frac{PN}{AT} = \frac{3}{2}$

Q.29 Answer:

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Solution:

Given equation of parabola is

$y^2 = 8x$

Let $\mathrm{P}\left(2\mathrm{t}^2, 4\mathrm{t}\right)$ **be the point of contact of tangent to the parabola. Equation of tangent at** P **is**

$\mathrm{ty} = \mathrm{x} + 2\mathrm{t}^2$

$\text{or } \mathrm{y} = \frac{\mathrm{x}}{\mathrm{t}} + 2\mathrm{t}$

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Equation of perpendicular line to tangent through focus $(2,0)$ is

$y=-t(x-2)$

$\Rightarrow \frac{x}{t}+2t=-t(x-2)$

$\Rightarrow x=0$

$\Rightarrow y=2t$

Coordinates of T are $(0,2t)$

Let (h,k) be the middle point of P and T

Then, $h=\frac{2t^2}{2}$ and $k=3t$

$\Rightarrow h=\left(\frac{k}{3}\right)^2$

$y^2=9x$

Length of latus rectum is 9.

Q.30 Answer:

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Solution:

Chord of the contact w.r.t. point $O(-1,2)$ is

$y=(x-1) \quad (\text{using} yy_1=2a(x+x_1))$

solving $y=x-1$, with parabola, we get the points of intersection as

$P(3+2\sqrt{2},2+2\sqrt{2})$ and $Q(3-2\sqrt{2},2-2\sqrt{2})$

$\therefore PQ^2=32+32=64$

$\therefore PQ=8$

Also length of perpendicular from $O(-1,2)$ on P Q

$=\frac{4}{\sqrt{2}}$

Then required area of triangle is

$A=\frac{1}{2}\cdot 8\cdot\left(\frac{4}{\sqrt{2}}\right)=8\sqrt{2}$ sq. units

$\frac{A}{\sqrt{2}}=8$

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